

Persistence of vaccine-elicited immune response up to 14 years post-HIV gp120-NefTat/AS01_B vaccination



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ABSTRACT

Background: Vaccines eliciting protective and persistent immune responses against multiple human immunodeficiency virus type 1 (HIV-1) clades are needed. This study evaluated the persistence of immune responses induced by an investigational, AS01-adjuvanted HIV-1 vaccine as long as 14 years after vaccination.

Methods: This phase I, open-label, descriptive, mono-centric, extension study with a single group (NCT03368053) was conducted in adults who received ≥ 3 doses of the clade B gp120-NefTat/AS01_B vaccine candidate 14 years earlier in a previous clinical trial (NCT00434512). Binding responses of serum antibodies targeting a panel of envelope glycoproteins, including gp120, gp140 and V1V2-scaffold antigens and representative of the antigenic diversity of HIV-1, were measured by binding antibody multiplex assay (BAMA). The gp120-specific CD4⁺/CD8⁺ T-cell responses were assessed by intracellular cytokine staining assay.

Results: At Year 14, positive IgG binding antibody responses were detected in 15 out of the 16 antigens from the BAMA V1V2 breadth panel, with positive response rates ranging from 7.1% to 60.7%. The highest response rates were observed for clade B strain V1V2 antigens, with some level of binding antibodies against clade C strains. Anti-V1V2 IgG3 response magnitude breadth, which correlated with decreased risk of infection in a previous efficacy trial, was of limited amplitude. Response rates to the antigens from the gp120 and gp140 breadth panels ranged from 7.7% to 94.1% and from 15.4% to 96.2% at Year 14, respectively. Following stimulation with gp120 peptide pool, highly polyfunctional gp120-specific CD4⁺ T-cells persisted up to Year 14, with high frequencies of CD40L tumor necrosis factor alpha (TNF- α), CD40L interleukin-2 (IL-2), CD40L TNF- α IL-2 and CD40L interferon gamma (IFN- γ) TNF- α IL-2 CD4⁺ T-cells, but no CD8⁺ T-cells detected.

Conclusions: Persistent antibodies binding to HIV-1 envelope glycoproteins, including the V1V2-scaffold, and gp120-specific cellular immunity were observed in volunteers vaccinated 14 years earlier with the gp120-NefTat/AS01_B vaccine candidate.

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Abbreviations: HIV, human immunodeficiency virus; HIV-1, HIV type 1; gp120, glycoprotein 120; Ig, immunoglobulin; V1V2, first and second variable regions; V3, third variable region; Fc, constant fragment; HVTN, HIV Vaccine Trials Network; NefTat, negative regulatory factor and trans-activator of transcription; gp120-NefTat vaccine, multi-antigen recombinant HIV-1 vaccine composed of clade B gp120 and a NefTat fusion protein; AS, adjuvant system; SHIV, simian/human immunodeficiency virus; IL-2, interleukin-2; CHO, Chinese hamster ovary; MPL, 3-O-desacyl-4'-monophosphoryl lipid A; QS21, *Quillaja saponaria* fraction 21; BAMA, binding antibody multiplex assay; CMI, cell-mediated immunity; ICS, intracellular cytokine staining; PBMC, peripheral blood mononuclear cell; CEVAC, Center for Vaccinology; MFI, mean fluorescence intensity; RNA, ribonucleic acid; CI, confidence interval; MB, magnitude breadth; AUC-MB, the area under the MB curve; CD40L, CD40 ligand; TNF- α , tumor necrosis factor alpha; IFN- γ , interferon gamma; GMR, geometric mean ratio; SAS, Statistical Analysis Systems; SAS SDD, SAS Drug Development; ADCC, antibody-dependent cellular cytotoxicity; ADPC, antibody-dependent cellular phagocytosis.

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1. Introduction

In 2018, 37.9 million people were living with human immunodeficiency virus (HIV) and 1.7 million new infections were estimated worldwide [1]. The availability of a safe, highly effective and accessible HIV vaccine would be a valuable complement to other preventive interventions, significantly contributing to the interruption of HIV transmission [2].

A previous study conducted in more than 16,000 participants in Thailand (RV144 study; NCT00223080) provided the first evidence of protective efficacy against HIV type 1 (HIV-1) acquisition; four priming injections of a recombinant canarypox vector vaccine (ALVAC-HIV [vCP1521]) and two booster injections of a bivalent recombinant clade B and E glycoprotein 120 (gp120) subunit vaccine using alum adjuvant, or a placebo, were administered in adults [3]. However, this protective effect rapidly declined one year after vaccination and vaccination did not appear to affect viral load in either early or late infection [4]. Nevertheless, immune-correlates analyses identified vaccine-induced immunoglobulin (Ig) G and IgG3 responses against the first and second variable regions (V1V2) of an HIV-1 gp120 envelope glycoprotein, IgG responses to the third variable region (V3) of HIV-1 gp120, CD4+ T-cell subsets, and antibody constant fragment (Fc) effector functions as being correlated with decreased HIV-1 risk, highlighting the fact that a complex interplay of immune responses contributed to the protective efficacy against HIV-1 acquisition [5–15].

Although the findings of the RV144 study paved the way to a preventive HIV vaccine using the prime-boost technology, immune responses needed to be strengthened and persistence improved. In this context, a phase I/II study conducted by the HIV Vaccine Trials Network (HVTN) program in South Africa evaluated an adapted vaccine regimen, using a recombinant canarypox vector vaccine expressing HIV-1 antigens from strains circulating in South Africa (ALVAC-HIV [vCP2438]) and a bivalent clade C gp120 subunit vaccine and MF59 adjuvant. Vaccination induced strong humoral and cellular immune responses, leading to qualification of this vaccine regimen for phase IIb/III efficacy testing [16]. Another investigational vaccine, the mosaic Ad26/Ad26+ gp140 HIV-1 vaccine, induced a robust immune response in healthy, HIV-uninfected participants, and is currently under evaluation in a phase IIb efficacy study (NCT02315703) in sub-Saharan Africa [17]. An additional question was whether a highly-immunostimulatory adjuvant could improve the vaccine-induced immune responses. Two phase I/IIa clinical trials are currently ongoing to evaluate the safety and immunogenicity of vaccine regimens including the DNA-HIV-PT123 vaccine (NCT02915016) or the ALVAC-HIV (vCP2438) vaccine (NCT03122223) in combination with the bivalent clade C gp120 vaccine candidate adjuvanted with MF59 or adjuvant system (AS)01_B in healthy, HIV-uninfected adults.

A multi-antigen recombinant HIV-1 vaccine composed of clade B gp120 and a negative regulatory factor and trans-activator of transcription (NefTat) fusion protein (gp120-NefTat vaccine) has been shown to protect rhesus monkeys against simian/HIV (SHIV)-induced disease [18]. In a phase I/II, double-blind, randomized controlled study, HIV-uninfected volunteers received four doses of the gp120-NefTat vaccine, adjuvanted with either AS02_A, AS02_V or AS01_B according to a 0, 1, 3 and 6 months schedule [19]. The vaccines used in the trial were overall well tolerated, and strong HIV-specific CD4+ T-cell responses, characterized by high lymphoproliferative capacity and interleukin-2 (IL-2) production, as well as strong binding antibody responses against gp120, Nef and Tat, were detected up to 18 months after the last vaccination. Since the strongest responses were observed with the AS01_B adjuvant, data on long-term persistent immune responses after vaccination with the gp120-NefTat/AS01_B vaccine would be helpful

to better characterize the role of this adjuvant in the immunogenicity of candidate HIV vaccines.

The objective of the present study was to evaluate the long-term persistence and breadth of binding antibody responses against V1V2, gp120 and gp140 specificities up to 14 years post-vaccination in HIV-uninfected volunteers who were vaccinated with the gp120-NefTat/AS01_B vaccine candidate (clade B, strain W61D) in the initial trial [19]. Other immune parameters, like the HIV-specific CD4+ T-cell and CD8+ T-cell responses, were also evaluated.

2. Methods

2.1. Study design

This phase I, open-label, descriptive, mono-centric, uncontrolled, extension study with a single group was conducted between 14 December 2017 and 30 January 2018 in selected individuals who participated in a previous randomized double-blind trial (NCT00434512) in Belgium. The design of the initial study has already been described in detail [19]. Briefly, healthy male and female HIV-seronegative volunteers aged 18–50 years were randomized into one of three groups to receive four doses (at 0, 1, 3 and 6 months) of study vaccine in one of the three different AS (AS02_A, AS02_V or AS01_B) and were followed up for 24 months post-first vaccination.

The vaccine antigens were prepared as a lyophilized pellet comprising 20 µg NefTat and 20 µg gp120 in sucrose, arginine and phosphate buffer [19]. NefTat comprises Nef, which is derived from HIV_{LAI} and Tat, which is derived from HIV_{BH10}. Nef and Tat are both expressed as a single fusion protein by recombinant technology in *Pichia pastoris* yeast cells. The recombinant gp120 is a truncated form of the envelope protein of the clade B HIV-1 isolate W61D produced in a Chinese hamster ovary (CHO) cell line. Prior to administration, the vaccine antigens were reconstituted with 0.5 mL of AS01_B, which is a GSK proprietary liposome-based AS containing 50 µg 3-O-desacyl-4'-monophosphoryl lipid A (MPL), 50 µg *Quillaja saponaria* fraction 21 (QS21; licensed by GSK from Antigenics LLC, a wholly owned subsidiary of Aenus Inc., a Delaware, USA corporation) and liposome.

This extension study consisted of a single sample collection visit for individuals having received gp120-NefTat adjuvanted with AS01_B, approximately 14 years post-vaccination. Blood samples were collected from all participants during the single study visit to assess humoral immunity by binding antibody multiplex assay (BAMA), and cell-mediated immunity (CMI) by intracellular cytokine staining (ICS) assays.

Retrospective serum samples that were collected in the initial study on Days 0 (pre-vaccination), 182 and 672 were re-tested for humoral immunity by BAMA with approval from the Duke University Institutional Review Board (Pro00100429). Retrospective peripheral blood mononuclear cell (PBMC) samples collected in the initial study on Days 0, 98 and 672 were re-tested for CMI responses using ICS assays.

This study was conducted at the Center for Vaccinology (CEVAC) in Ghent, Belgium. The study was approved by the local independent ethics committee and conducted in accordance with the Declaration of Helsinki and Good Clinical Practice guidelines. Written informed consent was obtained from all participants prior to study entry. The study is registered at <http://www.clinicaltrials.gov> (NCT03368053). The protocol is available at <http://www.gsk-clinicalstudyregister.com> (ID 201606). Anonymized individual participant data and study documents can be requested for further research at <https://www.clinicalstudydatarequest.com>.

2.2. Study objectives

The primary objectives were to determine the BAMA anti-V1V2 (IgG, IgG1, IgG2, IgG3, and IgG4) response rates and mean fluorescence intensity (MFI) response magnitude at Year 14 post-vaccination, and at historical time points of the initial study (Days 0, 182, and 672).

Secondary objectives were to determine the magnitude, response status, and cytokine co-expression profile of HIV-1 specific CD4⁺ T-cell and CD8⁺ T-cell responses as assessed by ICS assay at Year 14, and at historical time points of the initial study (Days 0, 98, and 672), and the anti-gp120 antibody response rates and magnitudes, as measured by BAMA, at Year 14, and at historical time points of the initial study (Days 0, 182, and 672). Anti-gp140 antibody response rates and magnitudes were also evaluated (exploratory objective).

2.3. Study population

Eligible participants were adults who had received at least three doses of the gp120-NefTat/AS01_B vaccine candidate in the initial study and would comply with the requirements of the protocol in the opinion of the investigator. Exclusion criteria included the use of any investigational or non-registered product during 30 days prior to study enrollment, HIV-1 or HIV-2 infection, participation to another clinical trial of an investigational HIV vaccine between the initial study and this extension study, immunosuppression from any cause (except administration of inhaled and topical steroids), administration of prednisone at a dosage of ≥ 20 mg/day, administration of cytotoxic medication within six months preceding this study, or of Igs or blood products within three months before enrollment, and history of vaccination with another investigational vaccine containing AS01. To exclude participants with HIV infection, the fourth-generation screening test to detect antibodies to HIV-1 and HIV-2 and p24 antigen (VIDAS DUO) was performed at the investigator's laboratory. In case of reactivity, an *Inno Lia* confirmation test (*Fujirebio*) was performed to distinguish between a real seropositivity and a vaccine-induced seropositivity.

2.4. Immunogenicity assessments

Custom BAMA was performed on serum samples at Duke University to evaluate the binding responses of serum antibodies targeting a panel of envelope glycoproteins, including gp120 and gp140, and V1V2-scaffold antigens. The BAMA breadth panel of cross-clade antigens included 16 V1V2 antigens, eight gp120 proteins and eight gp140 proteins, as previously described [20], representing different clades from diverse geographic regions and including circulating T/F strains [11,12,21,22]. Phylogenetic trees showing the alignment of the vaccine strain antigen gp120 (Clone W6.1D) with the antigens from the V1V2, gp120 and gp140 breadth panels are shown in [Supplementary Fig. 1](#). Binding antibody response rates and magnitudes were evaluated for total IgG for all 32 BAMA breadth panel antigens, and for the IgG3 subclass for the 16 V1V2 BAMA breadth panel antigens. In addition, binding antibody response rates and magnitudes were evaluated for total IgG and for the IgG1, IgG2, IgG3 and IgG4 subclasses for a set of six additional analytes, which were consensus, clade- or vaccine-matched antigens (1086C gp140C [also a part of the gp140 breadth panel], 1086C_D7gp120, C.1086C_V1_V2 Tags [referred to as C.1086C_V1V2], Con 6 gp120/B, Con S gp140 CFI, gp120 [Clone W6.1D]). HIV-1-specific total IgG was assessed at a 1:50 serum dilution, and IgG1–IgG4 subclasses at a 1:40 serum dilution in assay diluent, as previously described [21]. BAMA is a validated assay. Each sample was tested in duplicate within each assay with Quality Control checks for the agreement of the coefficient of

variation (%) with pre-set criteria [11,22,23]. The assay was repeated for samples that failed the first Quality Control check; if they failed again, the samples were excluded from the analyses. The unit for BAMA antibody level measurements is the median fluorescence intensity. For each sample, the mean of the two readings obtained from each duplicate was reported as MFI, referring to the final background- and blank bead-subtracted BAMA response magnitude. The linear range of binding magnitude at dilutions 1:50 and 1:40 was between 100 and 22,000 MFI. All assays were performed under Good Clinical Laboratory Practice compliance with antigen tracking (Levey-Jennings). CH58 monoclonal antibody [24] standard curve titrations were run in each assay and the concentration of V1V2 antibodies in the sera of vaccinees were calculated using 5-PL logistic regression (*BioPlex Manager*, v6.1) for positive samples within the linear range at the dilution tested.

HIV-1 specific CD4⁺ T-cells and CD8⁺ T-cells were evaluated by ICS on PBMCs at CEVAC Laboratory (Ghent, Belgium). The ICS experiments were performed in monoplicity, but samples collected at all the timepoints were tested in the same run for each participant. PBMCs were stimulated for two hours by 15 amino acid peptides with 11 amino acid overlaps covering the gp120 vaccine antigen (W61D clade B). After incubation with Brefeldin A and overnight stimulation, cells were stained with fluorescently labelled anti-CD4 and anti-CD8 markers. After permeabilization with saponin, cells were incubated with fluorescently labelled, cytokine-specific or activation marker-specific antibodies. Cytokine-producing cells were detected by flow cytometry. Results were presented as frequency of antigen-specific CD4⁺ or CD8⁺ T-cells expressing cytokine/marker upon *in vitro* stimulation.

2.5. Statistical analyses

Since 60 participants were vaccinated with the gp120-NefTat/AS01_B vaccine candidate in the primary study, approximately 30 to 40 participants were expected to be eligible for this long-term follow-up study. The total cohort included all participants who signed the informed consent and provided a blood sample. The per-protocol cohort for immunogenicity included all participants from the total cohort who met all eligibility criteria and presented a negative HIV ribonucleic acid (RNA) test at Year 14.

Demographic characteristics were summarized using descriptive statistics. Antigen- and subclass-specific response rates, as well as response magnitudes for binding antibodies for the V1V2, gp120 and gp140 breadth panel antigens and additional analytes, were calculated with 95% confidence intervals (CIs), at each time point, overall and in positive responders only. A positive responder was a participant with a positive sample at follow-up visits, i.e. having a response magnitude at least equal to the 95th percentile of all pre-vaccination MFI values or 100 MFI, whichever was greater, and with the ratio between sample-specific follow-up and pre-vaccination visit MFIs of at least three (before and after background subtraction). For samples missing pre-vaccination data, the median of available baselines was used to determine a positive response. The samples were considered as missing pre-vaccination data if no pre-vaccination sample was available for testing (5 samples), or if the pre-vaccination sample was available but failed the BAMA Quality Control checks after the first and repeat assay (0 to 3 samples depending on the antigen and isotype combination). Response magnitudes were presented as geometric mean of MFIs for all participants and in positive responders only. The geometric mean of MFI is the statistical measure summarizing the MFIs of all the samples analyzed for a particular timepoint, antigen and IgG subclass. Confidence intervals for geometric means were calculated where sample size was greater than one, with wide confidence intervals indicating small sample size and/or broad range of responses. Sample-specific and group-level

magnitude breadth (MB) curves were used to display the breadth of binding antibodies to the gp120, gp140 and V1V2 breadth panels, by IgG subclass, in samples with positive responses [20,25]. The area under the MB curve (AUC-MB) was used to compare the data over the three time-points, for the antigens included in the BAMA breadth panels. For samples with data for all antigens in the respective pre-defined breadth panels, breadth for a given subclass is the proportion of antigens in the given panel having the log10 MFI values greater than the values depicted on X-axis.

The following parameters were summarized using descriptive statistics at each time point where CMI results were available: frequency of gp120-specific CD4+/CD8+ T-cells expressing at least two markers among CD40 ligand (CD40L), IL-2, tumor necrosis factor alpha (TNF- α) and interferon gamma (IFN- γ), geometric mean ratios (GMRs) of frequency of gp120-specific CD4+/CD8+ T-cells expressing at least two markers among CD40L, IL-2, TNF- α and IFN- γ at each post-vaccination time point over pre-vaccination, and vaccine response rates for gp120-specific CD4+ T-cells expressing at least two markers among CD40L, IL-2, TNF- α and IFN- γ . Vaccine response rates correspond to the proportion of responders, defined as participants with a two-fold increase as compared to the cut-off for participants with pre-vaccination frequency below the cut-off ($\leq 0.0354\%$) and at least two-fold increase as compared to pre-vaccination frequency, for participants with pre-vaccination frequency above the cut-off ($> 0.0354\%$).

Statistical analyses were performed using the *Statistical Analysis Systems Drug Development (SAS DD)*. Plots and AUC-MB calculations were generated using *R Core Team 2018* [26].

3. Results

3.1. Study population

A total of 36 volunteers were enrolled in this follow-up study. Two participants were eliminated from the per-protocol cohort for immunogenicity: one due to lack of vaccination with gp120-NefTat/AS01_B, and one due to HIV infection at baseline.

The total cohort included 13 women and 23 men (Table 1). The mean (standard deviation) age of the study participants at Year 14 was 43.5 (8.6) years. Almost all participants were of Caucasian ancestry.

3.2. Humoral immunogenicity

3.2.1. V1V2 IgG binding responses

At Year 14, positive total IgG binding antibody responses were detected in 15 out of the 16 antigens from the BAMA V1V2 breadth panel. Different levels of anti-V1V2 total IgG response rates were observed depending on the HIV strain used in the assay, ranging from 43.3% to 100% at Day 182, from 3.2% to 83.3% at Day 672, and from 0.0% to 60.7% at Year 14 (Supplementary Table 1). The highest total IgG binding antibody response rates were observed for clade B strain V1V2 antigens, ranging from 43.3% to 100% at Day 182 and from 0.0% to 60.7% at Year 14. Anti-V1V2 total IgG response rate was above 50% for three out of the 16 antigens from the BAMA V1V2 breadth panel tested at Year 14: clade B gp70-700010058 V1V2, clade B gp70_B.CaseA V1V2 and clade C gp70-BF1266_431a V1V2 (Supplementary Table 1). Total IgG binding antibody response rates for clade C strain V1V2 antigens ranged from 73.3% to 100% at Day 182, and from 7.1% to 53.6% at Year 14.

Total IgG response magnitude for all antigens included in the BAMA V1V2 breadth panel were < 100 geometric mean of MFI at Day 0, ranged from < 100 to 21,648 geometric mean of MFI at Day 182, and < 100 to 1104 geometric mean of MFI at Day 672 in the initial study, compared with < 100 to 167 geometric mean

Table 1

Demographic characteristics (total cohort).

Characteristics	Value
Age at Year 14 (years)	
Mean (SD)	43.5 (8.6)
Median (Min–Max)	40.0 (32–61)
Gender	
Female, n (%)	13 (36.1)
Male, n (%)	23 (63.9)
Geographic Ancestry	
White/Caucasian heritage, n (%)	35 (97.2)
Black or African American heritage, n (%)	1 (2.8)

Max, maximum; Min, minimum; n (%), number (percentage) of participants in a given category; SD, standard deviation.

of MFI at Year 14 (Fig. 1 and Supplementary Table 1). In positive responders only, total IgG response magnitude for all antigens included in the BAMA V1V2 breadth panel ranged from 514 to 21,648 geometric mean of MFI at Day 182, and 777 to 5562 geometric mean of MFI at Day 672 in the initial study, compared to 517 to 2512 geometric mean of MFI at Year 14 (Supplementary Table 1). High MFIs at Year 14 were measured in samples taken from participants who were also among the highest responders at Day 182 and Day 672. The rank order was maintained, and most MFIs at Day 182 and Day 672 were above the upper limit of detection of the assay for samples with a high MFI at Year 14. The average AUC-MB for the vaccine-elicited anti-V1V2 total IgG binding antibody response decreased from 3.33 at Day 182 to 1.81 at Day 672 and to 0.95 at Year 14 (Supplementary Fig. 2).

Among the IgG subclasses tested for the 16 antigens from the BAMA V1V2 breadth panel, data were only available for IgG3, and responses were generally low and infrequent. At Day 182, Day 672 and Year 14, anti-V1V2 IgG3 response magnitude among all 16 antigens from the BAMA V1V2 breadth panel was <100 geometric mean of MFI (Fig. 2 and Supplementary Table 2). In positive responders only, IgG3 response magnitude for antigens included in the BAMA V1V2 breadth panel ranged from 104 to 323 geometric mean of MFI at Day 182, and 124 to 331 geometric mean of MFI at Day 672 in the initial study, compared with 249 to 263 geometric mean of MFI at Year 14 (Supplementary Table 2). The average AUC-MB for the vaccine-elicited anti-V1V2 IgG3 binding antibody response was 0.17 at Day 182, 0.03 at Day 672 and 0.01 at Year 14 (Supplementary Fig. 3).

3.2.2. gp120 and gp140 IgG binding responses

At Year 14, positive total IgG binding antibody responses were detected to all 16 antigens from BAMA gp120 and gp140 breadth panels.

The total IgG response rates to the eight antigens included in the BAMA gp120 breadth panel ranged from 78.1% to 100% at Day 182, from 51.6% to 100% at Day 672, and from 7.7% to 94.1% at Year 14 (Supplementary Table 3). Anti-gp120 total IgG response magnitude for the antigens in BAMA gp120 breadth panel ranged from 251 to 20,623 geometric mean of MFI at Day 182, from <100 to 3372 geometric mean of MFI at Day 672, and from <100 to 569 geometric mean of MFI at Year 14 for the different tested HIV strains and clades (Fig. 3 and Supplementary Table 3). In positive responders only, total IgG response magnitude for the gp120 breadth panel antigens ranged from 669 to 20,623 geometric mean of MFI at Day 182, and 170 to 4421 geometric mean of MFI at Day 672 in the initial study, compared with 122 to 965 geometric mean of MFI at Year 14 (Supplementary Table 3). The AUC-MB for the vaccine-elicited anti-gp120 total IgG binding antibody response decreased from 3.38 at Day 182 to 2.39 at Day 672 and to 1.51 at Year 14 (Supplementary Fig. 4).

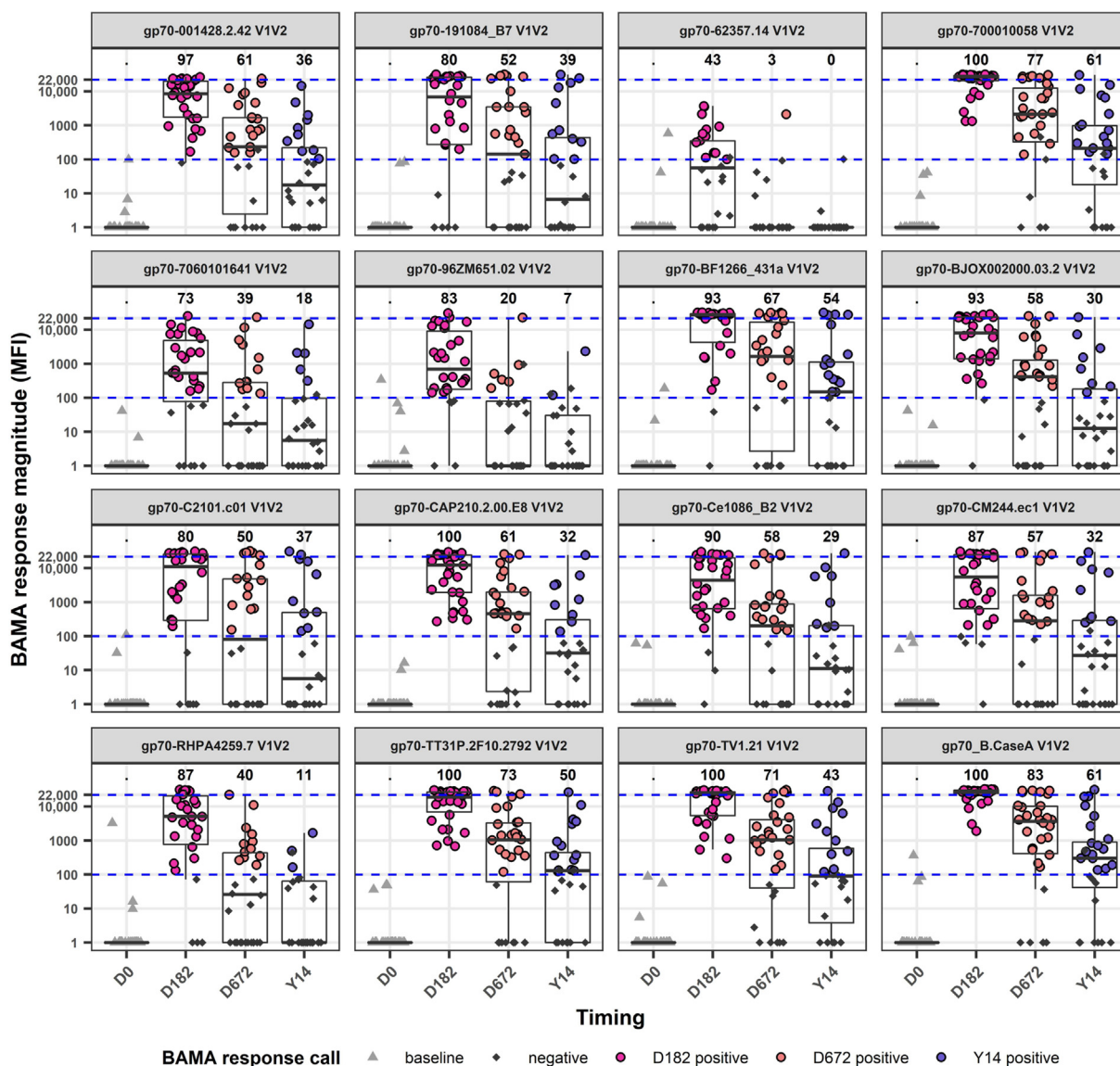


Fig. 1. Distribution of anti-V1V2 total IgG BAMA breadth panel response magnitude and response rates (per protocol cohort for immunogenicity). BAMA, binding antibody multiplex assay; D, Day; IgG, immunoglobulin G; MFI, mean fluorescence intensity; V1V2, first and second variable regions; Y, Year. Note: IgG binding magnitude for BAMA V1V2 breadth panel is shown for each sample by analyte and timing. The box-plot overlaps all samples, the mid-line of the box plot denotes the median, and the lower and upper edges denote the 25th and 75th percentiles, respectively. The dotted blue lines delimit the linear range of binding magnitude at dilution 1:50 [IgG]. The numbers on top of each box plot indicate the analyte- and timing-specific percent response rate, calculated as: number of positive samples/total number of samples $\times$ 100. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The total IgG response rates to the eight antigens included in the BAMA gp140 breadth panel ranged from 90.6% to 100% at Day 182, from 61.3% to 100% at Day 672, and from 15.4% to 96.2% at Year 14. Anti-gp140 total IgG response magnitude for the antigens in BAMA gp140 breadth panel ranged from 785 to 28,052 geometric mean of MFI at Day 182, from <100 to 16,538 geometric mean of MFI at Day 672, and from <100 to 2358 geometric mean of MFI at Year 14 for the different tested HIV strains and clades (Fig. 4 and Supplementary Table 3). In positive responders only, total IgG response magnitude for gp140 antigens ranged from 1565 to 28,052 geometric mean of MFI at Day 182, and 285 to 16,538 geometric mean of MFI at Day 672 in the initial study, compared with 187 to 3217 geometric mean of MFI at Year 14 (Supplementary Table 3). The AUC-MB for the vaccine-elicited anti-gp140 IgG binding antibody response decreased from 3.86 at Day 182 to 2.75 at Day 672 and to 1.85 at Year 14 (Supplementary Fig. 5).

For a given subclass and breadth panel combination (V1V2, gp120 and gp140), the sample-specific and group level MB curves are represented in Supplementary Figs. 2–5, and sample-specific AUC-MB values are represented in Supplementary Fig. 6.

3.2.3. Binding responses to the additional analytes

The BAMA responses for the six additional analytes run for all four subclasses were primarily IgG- and IgG1-directed, followed by IgG3. They peaked at Day 182 and their magnitude dropped gradually over time (Fig. 5). Limited IgG4 responses were also observed; no IgG4 response to C.1086C_V1V2 was detected. Most of the IgG2 responses were below the linear range of binding magnitude at dilution 1:40 (<100 MFI).

3.2.4. Binding responses to CH58 monoclonal antibody

The geometric mean for levels of CH58-like V1V2 IgG in participant serum ranged from 58.7 μ g/ml (95% CI: 46.2–74.7) at Day

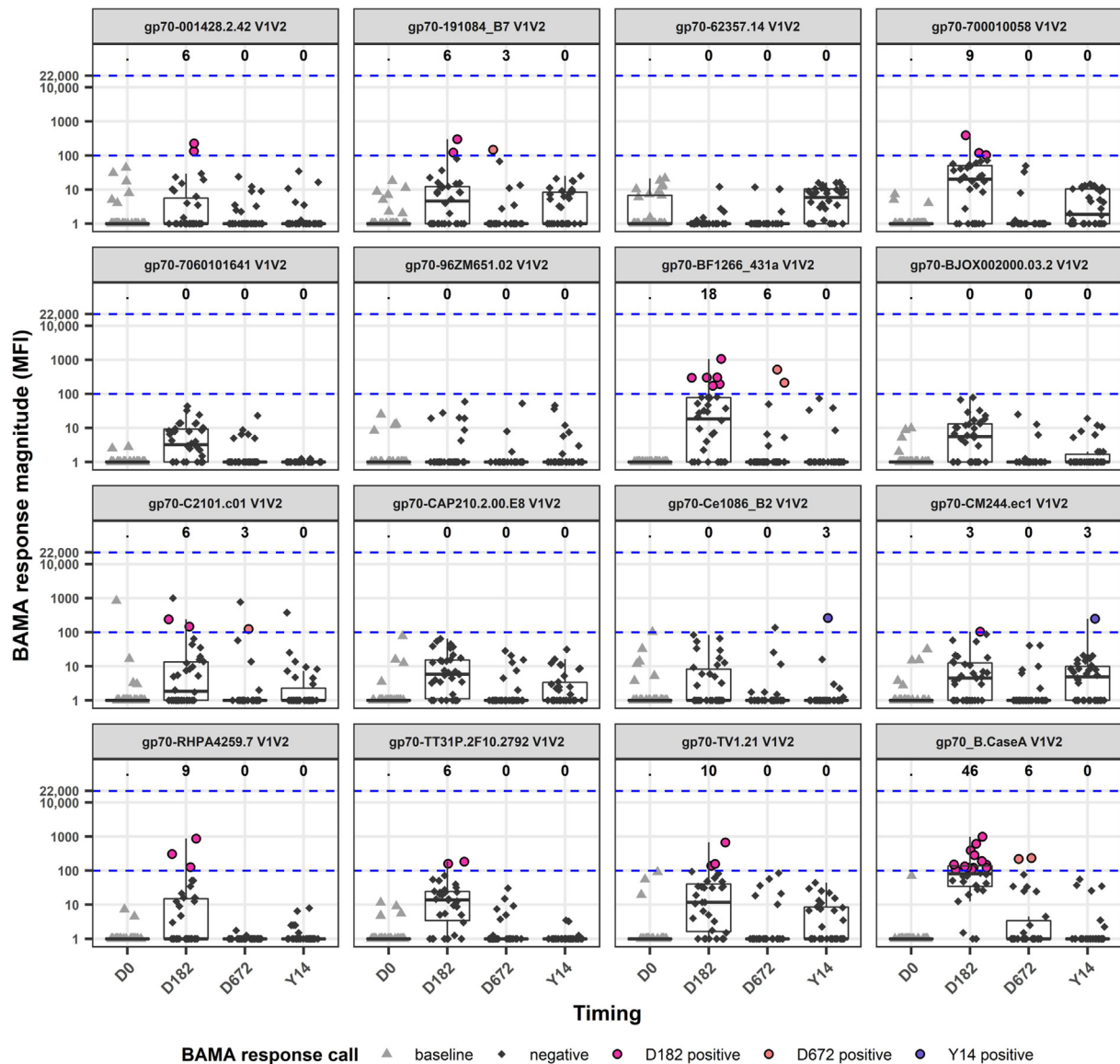


Fig. 2. Distribution of anti-V1V2 IgG3 BAMA breadth panel response magnitude and response rates (per protocol cohort for immunogenicity). BAMA, binding antibody multiplex assay; D, Day; IgG, immunoglobulin G; MFI, mean fluorescence intensity; V1V2, first and second variable regions; Y, year. Note: IgG3 binding magnitude for BAMA V1V2 breadth panel is shown for each sample by analyte and timing. The box-plot overlaps all samples, the mid-line of the box plot denotes the median, and the lower and upper edges denote the 25th and 75th percentiles, respectively. The dotted blue lines delimit the linear range of binding magnitude at dilution 1:40 [IgG3]. The numbers on top of each box plot indicate the analyte- and timing-specific percent response rate, calculated as: number of positive samples/total number of samples $\times$ 100. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

182, 13.5 $\mu\text{g/ml}$ (95%CI: 7.83–23.2) at Day 672 and 4.62 $\mu\text{g/ml}$ (95% CI: 2.02–10.6) at Year 14. Among the positive responders for whom concentration was in the linear range of assay, 100% of participants had levels of CH58-like IgG in serum $>2.98 \mu\text{g/ml}$ at Day 182, 80% at Day 672 and 47% at Year 14 (Supplementary Fig. 7).

3.3. Cellular immunogenicity

Following stimulation with the gp120 peptide pool (W61D clade B – vaccine strain), the mean frequency of gp120-specific CD4⁺ T-cells expressing at least two markers among CD40L, IL-2, TNF- α and IFN- γ was 0.007% at Day 0, 0.565% at Day 98, 0.376% at Day 672 and 0.272% at Year 14 (Fig. 6 and Supplementary Table 4). GMRs of frequency of gp120-specific CD4⁺ T-cells expressing at least two markers among CD40L, IL-2, TNF- α and IFN- γ at post-vaccination over pre-vaccination time points were 190.6 (95% CI:

74.9–485.4) at Day 98, 142.4 (95% CI: 52.5–386.8) at Day 672 and 105.9 (95% CI: 42.3–265.4) at Year 14. The vaccine response rate for gp120-specific CD4⁺ T-cells expressing at least two markers among CD40L, IL-2, TNF- α and IFN- γ were 96.2% (25/26) at Day 98, 100% (23/23) at Day 672, and 91.7% (22/24) at Year 14.

Highly polyfunctional gp120-specific CD4⁺ T-cells were induced and persisted, with high frequencies of CD40L TNF- α , CD40L IL-2, CD40L TNF- α IL-2 and CD40L IFN- γ TNF- α IL-2 CD4⁺ T-cells detectable up to 14 years post-primary vaccination (Fig. 7). Interestingly, CD4⁺ T-cells expressing the four markers were less frequently induced but had the strongest persistence.

In line with observations of the initial study, no CD8⁺ T-cells response were detected (Supplementary Table 4). GMRs of frequency of gp120-specific CD8⁺ T-cells expressing at least two markers among CD40L, IL-2, TNF- α and IFN- γ at post-vaccination over pre-vaccination time points were 0.9 (95% CI: 0.3–3.1) at

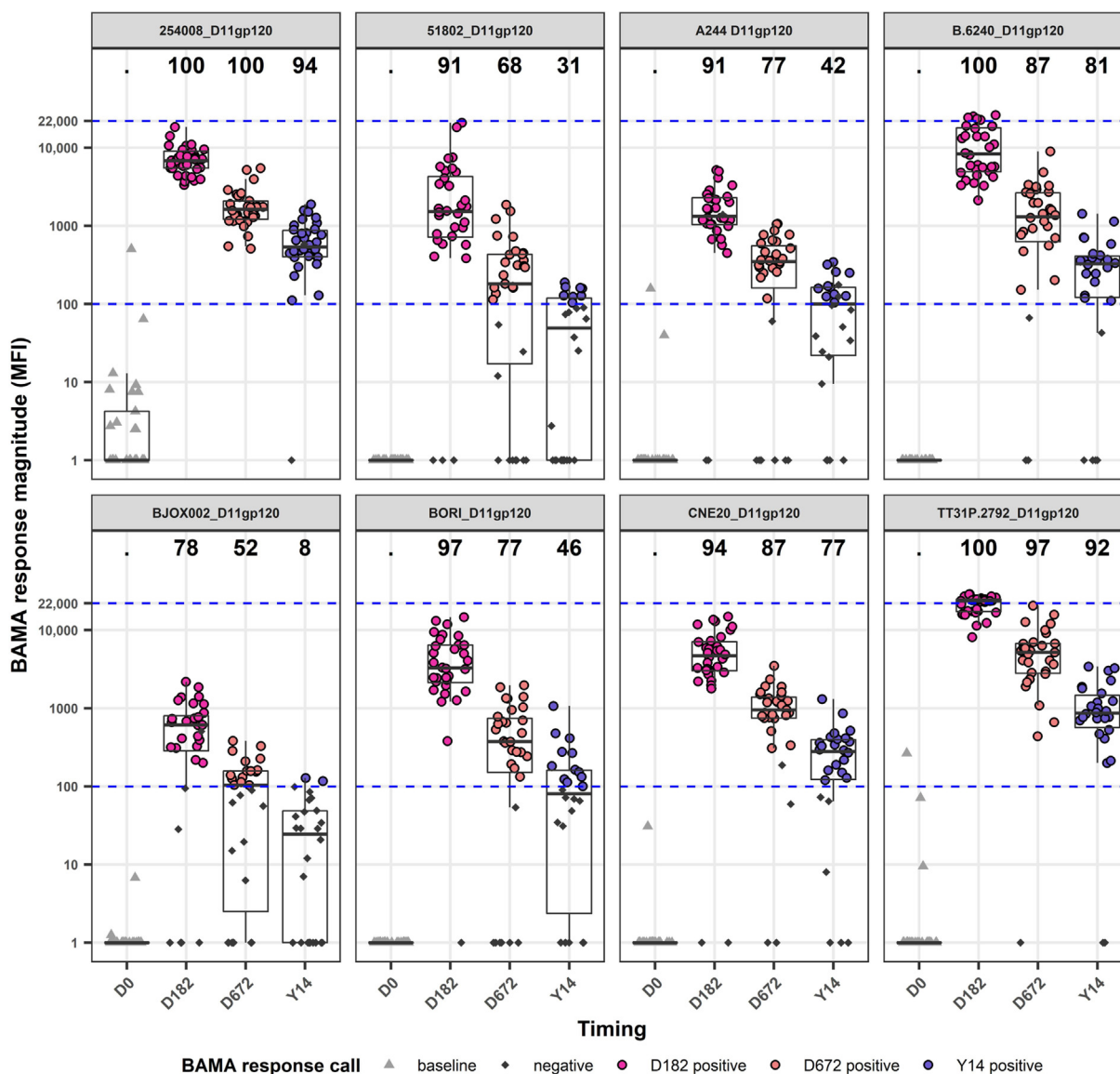


Fig. 3. Distribution of anti-gp120 total IgG BAMA breadth panel response magnitude and response rates (per protocol cohort for immunogenicity). BAMA, binding antibody multiplex assay; D, Day; gp120, glycoprotein 120; IgG, immunoglobulin G; MFI, mean fluorescence intensity; Y, year. Note: IgG binding magnitude for the BAMA gp120 breadth panel is shown for each sample by analyte and timing. The box-plot overlaps all samples, the mid-line of the box plot denotes the median, and the lower and upper edges denote the 25th and 75th percentiles, respectively. The dotted blue lines delimit the linear range of binding magnitude at dilution 1:50 [IgG]. The numbers on top of each box plot indicate the analyte- and timing-specific percent response rate, calculated as: number of positive samples/total number of samples $\times$ 100. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Day 98, 1.9 (95% CI: 0.5–6.8) at Day 672 and 1.6 (95% CI: 0.5–5.5) at Year 14.

4. Discussion

Prophylactic vaccines against HIV eliciting persistent and protective immune responses with substantial cross-clade coverage are needed [27,28]. In the only study to date showing an impact of vaccination on the risk of infection (RV144 study), vaccine efficacy of four priming injections of the ALVAC-HIV (vCP1521) vaccine and two booster injections of a bivalent recombinant alum-adsorbed gp120 subunit vaccine in adults was about 60% during the first year after initial vaccination but declined to about 30% at 42 months [4]. One approach to improve antibody persistence is to use antigen/adjuvant combinations that would elicit increased and durable responses [27]. The present study evaluated the persistence of the humoral binding immunity to envelope gly-

coproteins and of the cellular immunity in volunteers who received the gp120-NefTat/AS01_B vaccine candidate 14 years earlier [19].

First, we evaluated binding humoral responses against a panel of virus envelope glycoproteins, which are representative of the antigenic diversity of HIV-1 and included diverse sequences of the V1V2 region of gp120 [20]. The level of binding IgG antibodies to the panel of antigens was identified as a correlate of reduced risk of HIV-1 infection in the RV144 study [11,12,15,20,22]. In our trial, anti-V1V2 binding IgG antibodies for 15 out of the 16 antigens from the BAMA V1V2 breadth panel persisted up to 14 years post-vaccination. Binding antibody response rates (range: 0.0–60.7%) varied depending on the HIV strains, and the mean AUC-MB of binding antibody response decreased with time (from 3.33 at Day 182 to 0.95 at Year 14). Response magnitudes were low after 14 years compared to Day 182 and Day 672, with antigen specific geometric mean of MFIs ranging from below the linear

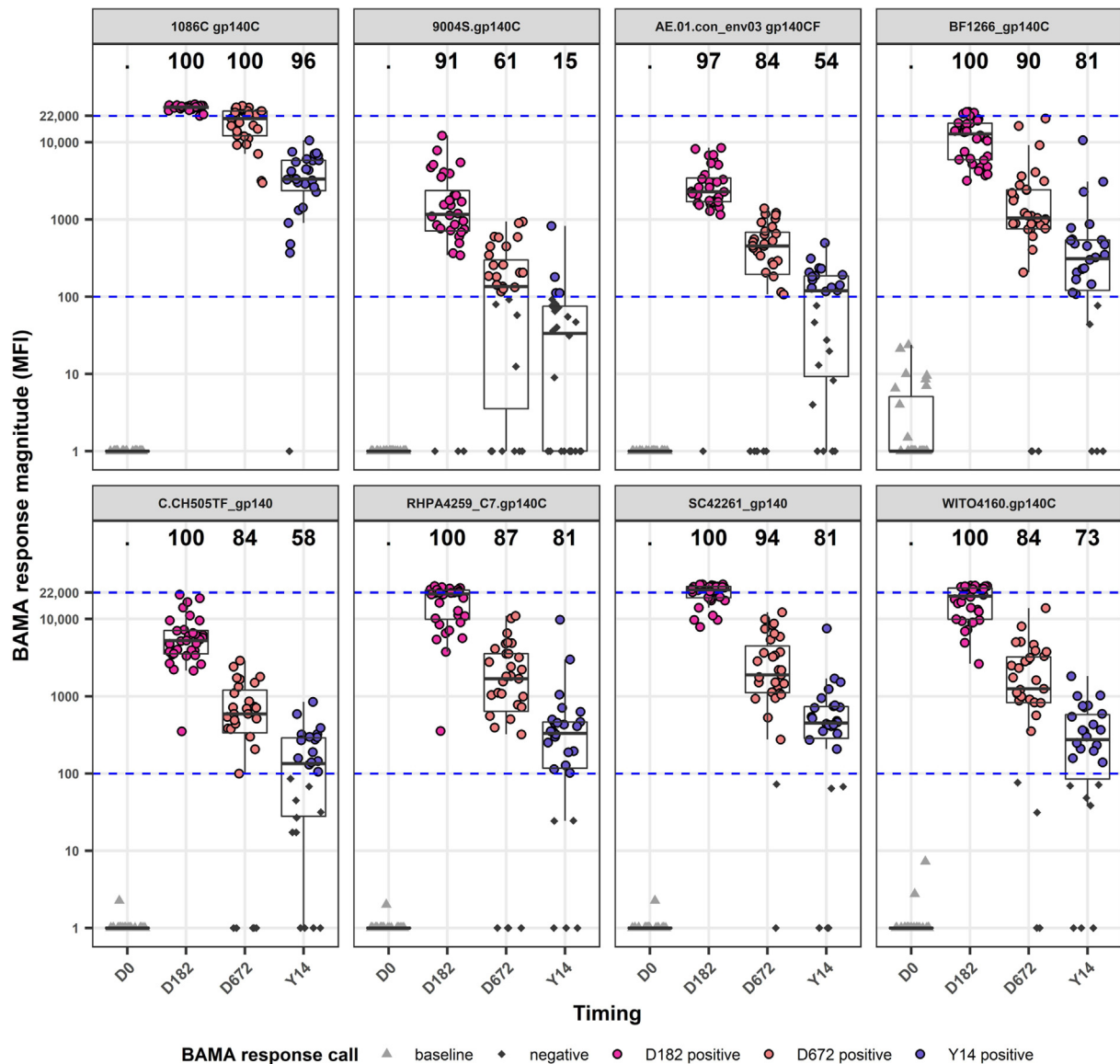


Fig. 4. Distribution of anti-gp140 total IgG BAMA breadth panel response magnitude and response rates (per protocol cohort for immunogenicity). BAMA, binding antibody multiplex assay; D, Day; gp140, glycoprotein 140; IgG, immunoglobulin G; MFI, mean fluorescence intensity; Y, Year. Note: IgG binding magnitude for the BAMA gp140 breadth panel is shown for each sample, by analyte and timing. The box-plot overlaps all samples, the mid-line of the box plot denotes the median, and the lower and upper edges denote the 25th and 75th percentiles, respectively. The dotted blue lines delimit the linear range of binding magnitude at dilution 1:50 [IgG]. The numbers on top of each box plot indicate the analyte- and timing-specific percent response rate, calculated as: number of positive samples/total number of samples $\times$ 100. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

range of the assay (<100 MFI) to 167 among all samples, and from 517 to 2512 among the positive responders. The highest responses were observed for clade B strain antigens, which was an expected finding since the strain used in the gp120-NefTat/AS01_B vaccine candidate was W61D (clade B). Moreover, there was also some level of cross-reactive binding antibodies against clade C strains. Concentrations of circulating CH58-like V1V2 IgG antibodies of at least 2.98 μ g/ml were previously reported to be associated with a decrease in HIV-1 infection in the RV144 HIV vaccine trial [14]. In our study, among the BAMA positive responders, 100% of vaccinees had CH58-like IgG serum concentrations >2.98 μ g/ml at Day 182, 80% at Day 672, and 47% at Year 14, indicating enhanced durability of vaccine-elicited V1V2 IgG levels associated with decreased risk of HIV-1 infection [14].

Persistence of anti-V1V2 binding antibodies of the IgG3 subclass was low against the various antigens tested, with poor overall response magnitude in all participants and ranging from 249 to

263 in positive responders only. Lower HIV-specific IgG3 responses were previously observed with repeated protein vaccination [12,20]. V1V2 IgG3 antibody responses were induced by ALVAC-HIV and a bivalent recombinant clade B and E gp120 subunit vaccine adjuvanted with alum in the RV144 trial, which correlated with a decreased risk of infection and were associated with antibody Fc effector functions, together with IgG1 responses [5,12,15,29]. Vaccination regimens combining primary doses of recombinant canarypox vector vaccine expressing HIV-1 antigens (ALVAC-HIV) and booster doses of AS01_B-adjuvanted gp120 protein could induce persistent immune responses with the desired antibody profile. This is currently evaluated in an ongoing clinical trial comparing the safety and immunogenicity of ALVAC-HIV and MF59- or AS01_B-adjuvanted bivalent subtype C gp120 in healthy, HIV-uninfected adults (NCT03122223).

Immune responses against anti-gp120 and -gp140 antigens from the panel of virus envelope glycoproteins also add useful

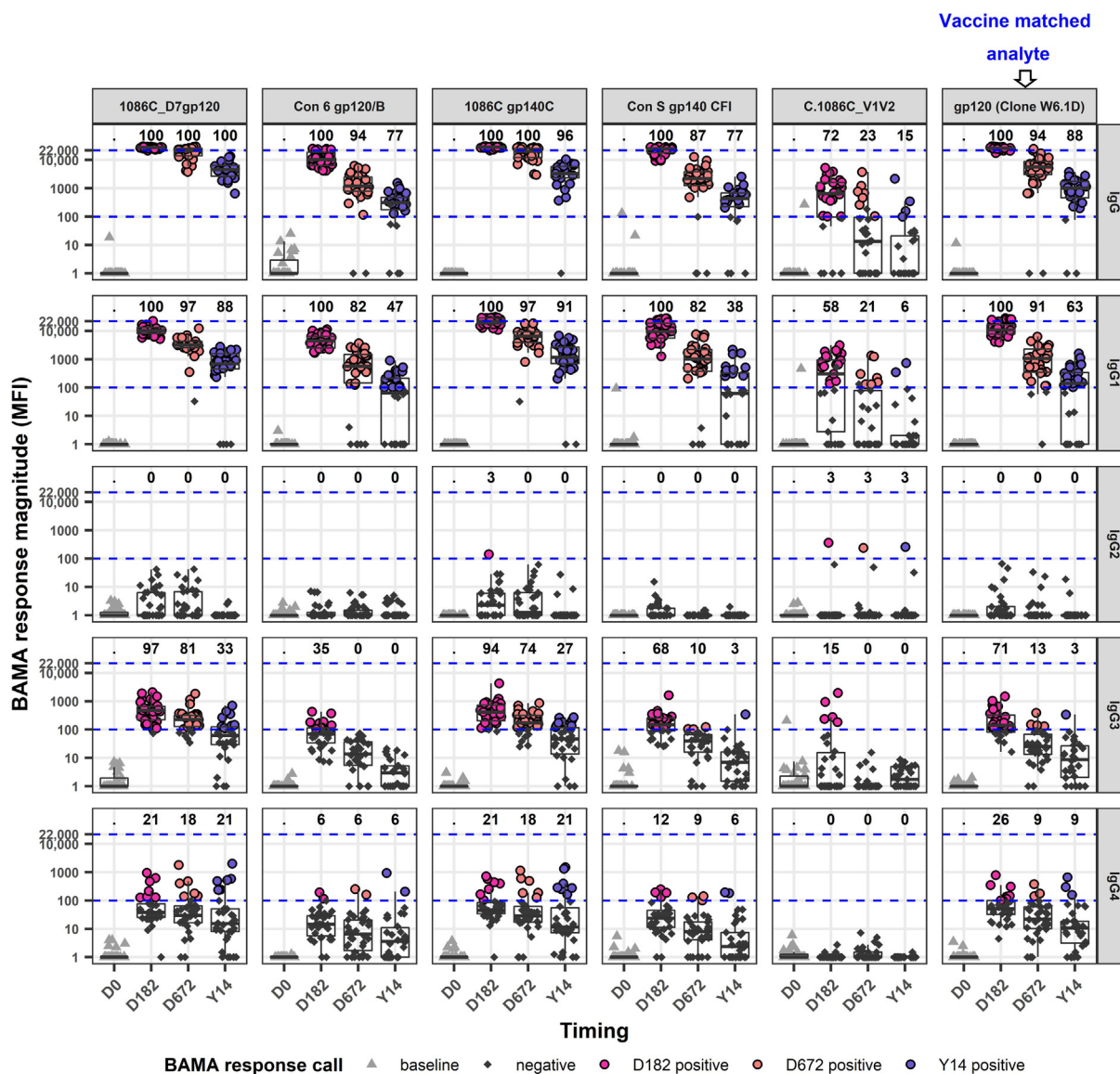


Fig. 5. Distribution of total IgG, IgG1, IgG2, IgG3 and IgG4 BAMA response magnitude and response rates for the six additional analytes (per protocol cohort for immunogenicity). BAMA, binding antibody multiplex assay; D, Day; IgG, immunoglobulin G; MFI, mean fluorescence intensity; Y, Year. The vaccine strain antigen [gp120 (Clone W6.1D)] is highlighted in the figure. Note: IgG, IgG1, IgG2, IgG3, IgG4 binding magnitude for BAMA is shown for each sample, by analyte and timing. The box-plot overlies all samples, the mid-line of the box plot denotes the median, and the lower and upper edges denote the 25th and 75th percentiles, respectively. The dotted blue lines delimit the linear range of binding magnitude at dilution 1:50 [IgG] and 1:40 [IgG1, IgG2, IgG3, IgG4]. The numbers on top of each box plot indicate the analyte- and timing-specific percent response rate, calculated as: number of positive samples/total number of samples $\times$ 100. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

information on binding diversity and on epitopes that are not targeted by broadly neutralizing antibodies but are sites of vulnerability [20]. Almost all vaccine recipients developed IgG binding antibodies to gp120 and gp140 antigens from various circulating strains of HIV-1, which were maintained up to 14 years post-vaccination with mean AUC-MB of 1.51 and 1.85, respectively. Response magnitudes for all antigens dropped gradually between Day 182 (range: 251–28,052 geometric mean of MFI) and 14 years post-vaccination (range: <100–2358 geometric mean of MFI). In addition to the binding humoral responses against the panel of virus envelope glycoproteins evaluated in our study, we cannot exclude the possibility that four doses of the gp120-NefTat/AS01_B vaccine induced other protective antibody subclasses and functionalities that are associated with increased protection, compared to the four priming injections of the ALVAC-HIV (vCP1521) vaccine and the two booster injections of alum-adsorbed gp120 as used

in the RV144 study. Continued investigation of vaccine efficacy should focus on the identification of both epitope-specific and functional antibody responses against circulating strains of HIV-1.

We also evaluated the humoral binding responses across IgG subclasses for six consensus, clade- or vaccine-matched antigens. These analytes included the clade C.1086 V1V2 antigen, an envelope protein immunogen used in efficacy studies [12], and the gp120 (Clone W6.1D) antigen, which is representative of the vaccine antigen. We showed that the BAMA responses induced by the gp120-NefTat/AS01_B vaccine up to 14 years post-vaccination were primarily IgG and IgG1-directed, with some IgG3 and limited IgG4 responses. The poor IgG3 response may be due to a lower abundance of IgG3 producing plasma cells elicited by the vaccine. A previous study in HIV-infected infants indicated that antibody-dependent cell-mediated cytotoxicity (ADCC) positively correlated with the magnitude of IgG1 binding and IgG1 levels were associ-

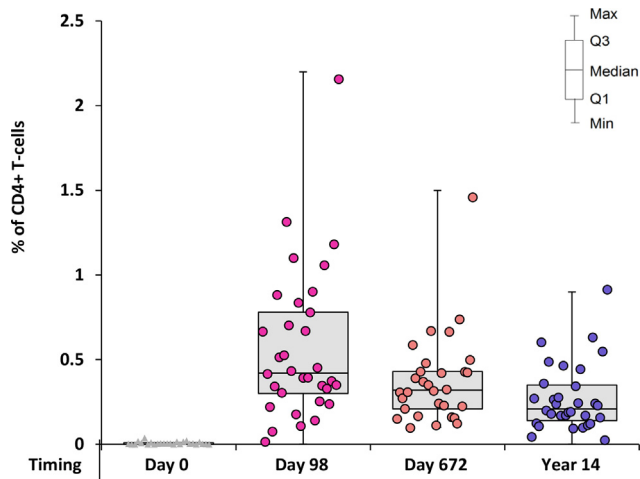


Fig. 6. Frequency of HIV-1 specific CD4⁺ T-cells expressing at least two markers among CD40 ligand (CD40L), interleukin-2 (IL-2), tumor necrosis factor alpha (TNF- α) and interferon gamma (IFN- γ) (per protocol cohort for immunogenicity). Note: Each dot represents sample-specific frequency of HIV-1 specific CD4⁺ T-cells expressing at least two markers among CD40 ligand (CD40L), interleukin-2 (IL-2), tumor necrosis factor alpha (TNF- α) and interferon gamma (IFN- γ), by timing. HIV-1, human immunodeficiency virus type 1; Max, maximum; Min, minimum; Q1, lower quartile; Q3, upper quartile.

ated with survival [30]. In another study, IgG1 monoclonal antibodies isolated from ALVAC/AIDS VAX vaccinees were shown to mediate ADCC activity [31]. High levels of both anti-V1V2 IgG3 and IgG1 antibodies led to enhanced antibody-dependent cellular phagocytosis (ADCP) in ALVAC-HIV (vCP1521)/clade B/E gp120 vaccinees [5]. Nevertheless, superior ADCP-mediated activity was reported for IgG3 monoclonal antibodies against multiple HIV-1 epitopes, including V2, compared to IgG1 antibodies [32]. Although the fact that IgG1-directed BAMA responses induced by the gp120-NefTat/AS01_B vaccine were still detected 14 years post-vaccination is a positive result even if the levels were low, additional analyses would be needed to draw any conclusion concerning the functional properties of the vaccine-induced antibodies.

Persistence of CD4⁺ T-cell responses against gp120 was observed in volunteers vaccinated with the gp120-NefTat/AS01_B vaccine candidate 14 years earlier, with a mean frequency of 0.272% of gp120-specific CD4⁺ T-cells expressing at least two markers among CD40L, IL-2, TNF- α and IFN- γ . Following stimulation by gp120, only gp120-specific CD4⁺ T-cells and no gp120-specific CD8⁺ T-cells were detected, and the CD4⁺ T-cells expressed mainly IL-2, although some cells also expressed TNF- α , which is in line with the results of the initial study [19]. This is an important finding since CD4⁺ T-cells may induce effective HIV immunity [33] and polyfunctional antigen-specific CD4⁺ T-cells could play a role in vaccine efficacy [13]. However, we did not evaluate in depth polyfunctional populations of this CD4⁺ T-cell phenotype, perhaps also expressing IL-4, which have been strongly associated with reduced risk of HIV-1 acquisition [13].

This was the first study to evaluate the long-term persistence of immune responses, particularly the IgG V1V2 immune correlate of HIV-1 risk, induced by an investigational vaccine against HIV-1 up to 14 years post-vaccination. The limitations of this study included the absence of control group, the small sample size, the absence of efficacy data, and the fact that specific CD4⁺ T-cell expressing TNF- α , IFN- γ , IL-4, IL-2 and CD40L were not evaluated. A further drawback was the absence of evaluation of the binding response to the V3 region of HIV-1 gp120, which was identified as another correlate of lower infection risk in the RV144 study [6].

5. Conclusions

To our knowledge, this is the first study to report such long-term humoral and cellular persistence after administration of an adjuvanted HIV vaccine. Persistence of antibodies binding to envelope glycoproteins and gp120-specific cellular immunity was observed in volunteers vaccinated with the gp120-NefTat/AS01_B vaccine candidate 14 years earlier. In particular, the persistence of antibodies binding to the V1V2 scaffold is encouraging, even if positive responses for IgG3 breadth panel antigens were limited. The persistence and the polyfunctionality of the gp120-specific CD4⁺ T-cell response, which was also identified as a correlate of

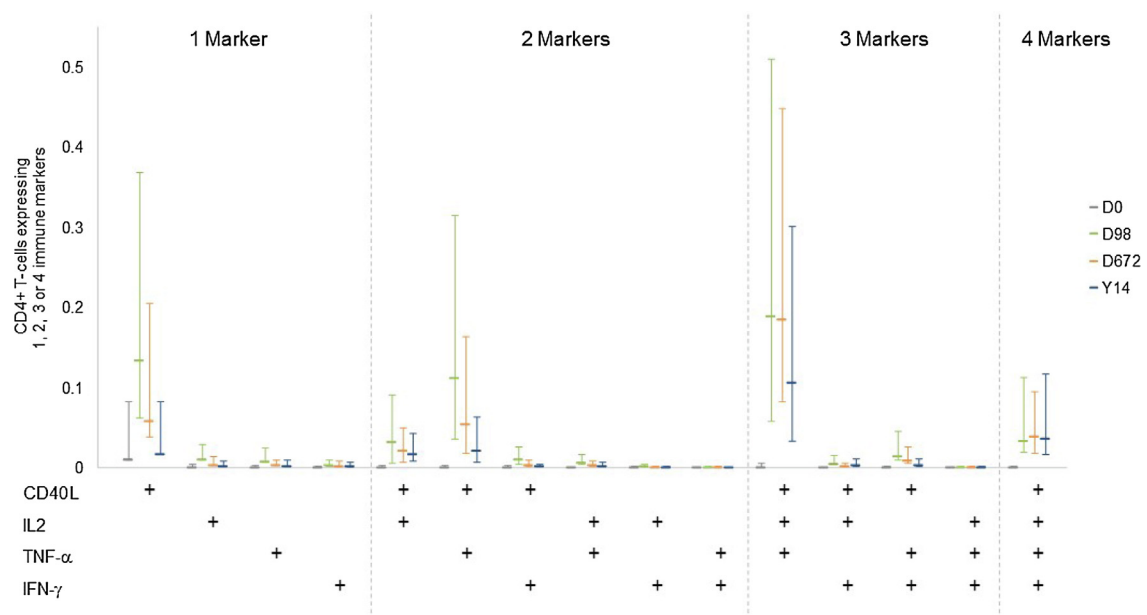


Fig. 7. Polyfunctional profile of the HIV-1 specific CD4⁺ T-cells for each combination of the 4 markers CD40L, IL-2, TNF- α and IFN- γ (per protocol cohort for immunogenicity). CD40L, CD40 ligand; D, Day; HIV-1, human immunodeficiency virus type 1; IL-2, interleukin-2; IFN- γ , interferon gamma; TNF- α , tumor necrosis factor alpha; Y, Year. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

lower risk of HIV infection in the RV144 study, were also remarkable. Our results suggest that the use of an adjuvant has an impact on the persistence of immune responses induced by HIV vaccines. This is indeed the first evidence that an AS01 adjuvanted HIV candidate vaccine can induce a persistent adaptive immune response, with HIV-specific circulating antibodies and immune cells maintained in subjects having received at least three doses of the vaccine 14 years earlier. These data add to recently published results showing that clade B envelope immunogens with MF59 adjuvant can elicit durable memory T- and B-cell responses, as demonstrated by recall booster vaccination [34].

6. Trademarks

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7. Financial disclosure

GlaxoSmithKline Biologicals SA sponsored this study/research and was involved in all stages of study conduct, including analysis of the data. GlaxoSmithKline Biologicals SA also took in charge all costs associated with the development and publication of this manuscript.

CRedit authorship contribution statement

Olivier Van Der Meeren: Methodology, Investigation, Formal analysis, Writing - review & editing. **Erik Jongert:** Methodology, Formal analysis, Writing - review & editing. **Kelly E. Seaton:** Methodology, Investigation, Formal analysis, Writing - review & editing. **Marguerite Koutsoukos:** Methodology, Investigation, Formal analysis, Writing - review & editing. **Annelies Aerssens:** Investigation, Formal analysis, Writing - review & editing. **Caroline Brackett:** Investigation, Formal analysis, Writing - review & editing. **Muriel Debois:** Formal analysis, Writing - review & editing. **Michel Janssens:** Investigation, Formal analysis, Writing - review & editing. **Geert Leroux-Roels:** Methodology, Investigation, Formal analysis, Writing - review & editing. **Doris Mesia Vela:** Formal analysis, Writing - review & editing. **Sheetal Sawant:** Investigation, Formal analysis, Writing - review & editing. **Nicole L. Yates:** Methodology, Investigation, Formal analysis, Writing - review & editing. **Georgia D. Tomaras:** Methodology, Investigation, Formal analysis, Writing - review & editing. **Isabel Leroux-Roels:** Methodology, Investigation, Formal analysis, Writing - review & editing. **François Roman:** Methodology, Investigation, Formal analysis, Writing - review & editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: [O.V.D.M., E.J., M.K., M.D., M.J., D.M.V. and F.R. are employed by the GSK group of companies. O.V.D.M., E.J., M.K., D.M.V. and F.R. also hold shares from the GSK group of companies. M.K. also reports receiving a grant from the European Commission for the EbolaVac program outside of the submitted work. K.S., C.B., S.S., G.D.T. received financial and other support without any personal financial benefit from the GSK group of companies for the conduct of the study. They also report grants from the GSK group of companies to their institution outside of the submitted work. A.A., G.L.R. and I.L.R. report grant from the GSK group of

companies to their institutions (Ghent University Hospital and Ghent University) to compensate the costs for the conduct of the study. I. LR. also reports grant from the GSK group of companies to her institutions (Ghent University Hospital and Ghent University) to finance the conduct of other vaccine trials. N.L.Y. has nothing to disclose.].

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.vaccine.2019.12.058>.

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